Community Organization

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with

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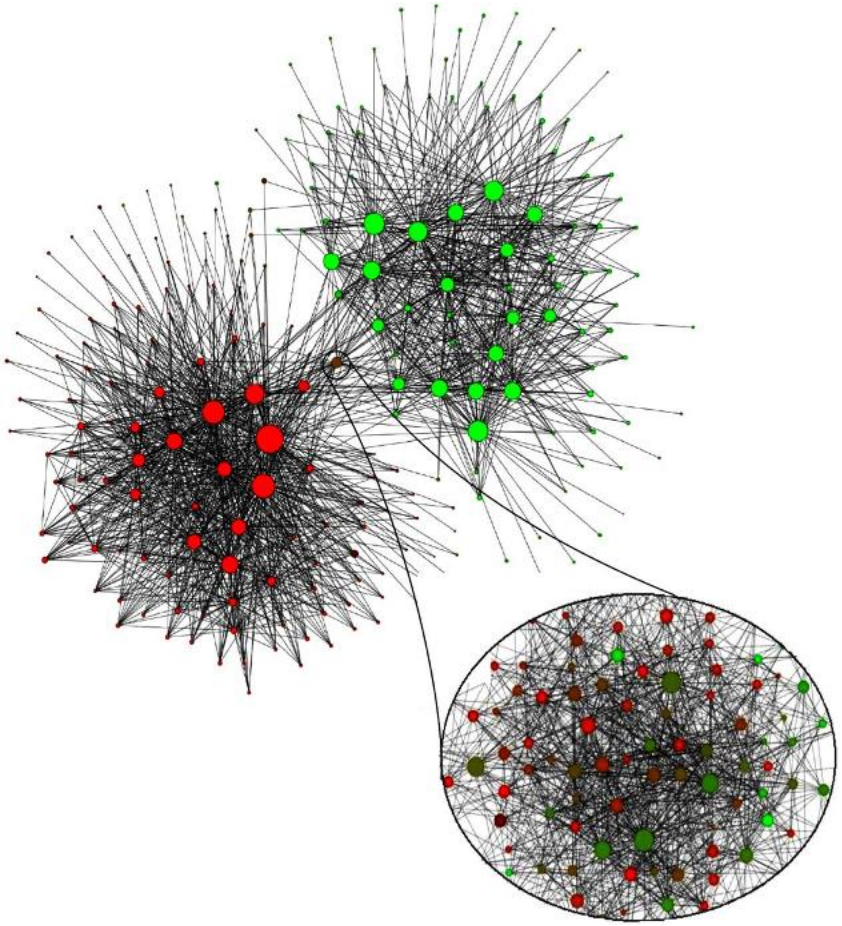
Slides/visuals: Mauro Martino

www.BarabasiLab.com

Introduction

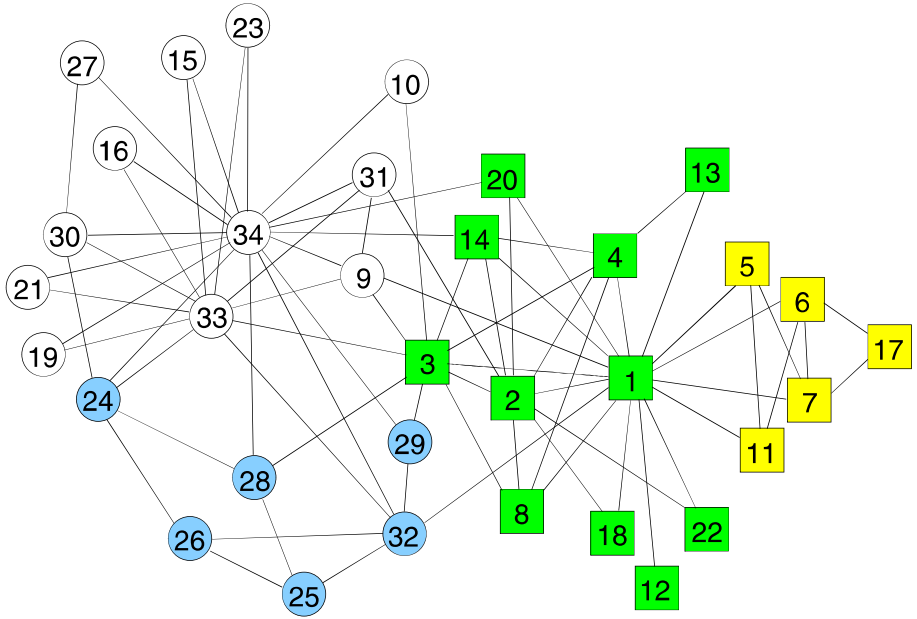
Section 1 Introduction



Examples of communities

Zachary's Karate Club

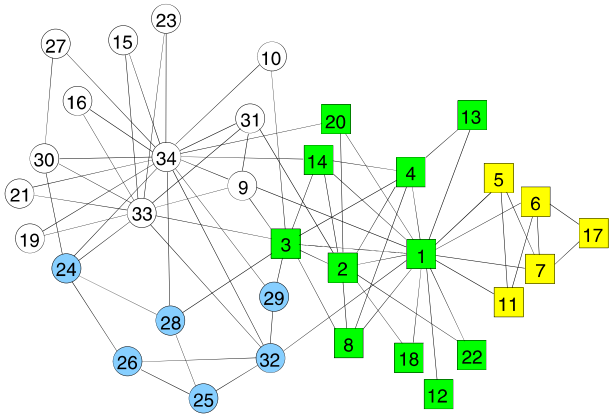
a

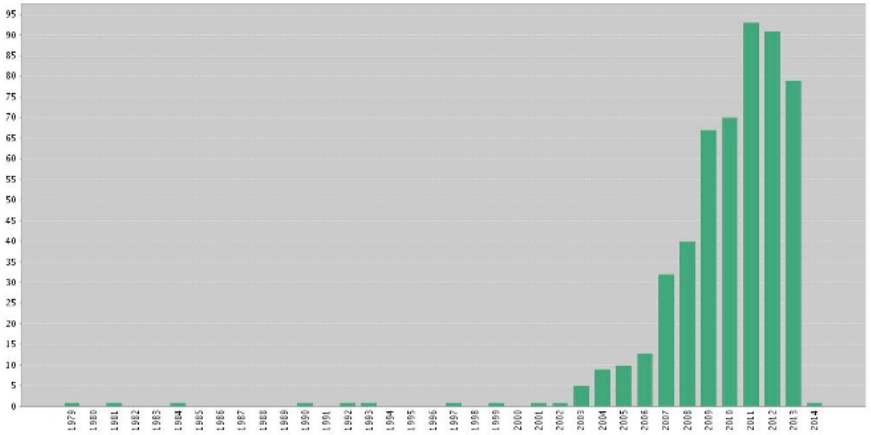


Section 2

Zachary's Karate Club

a

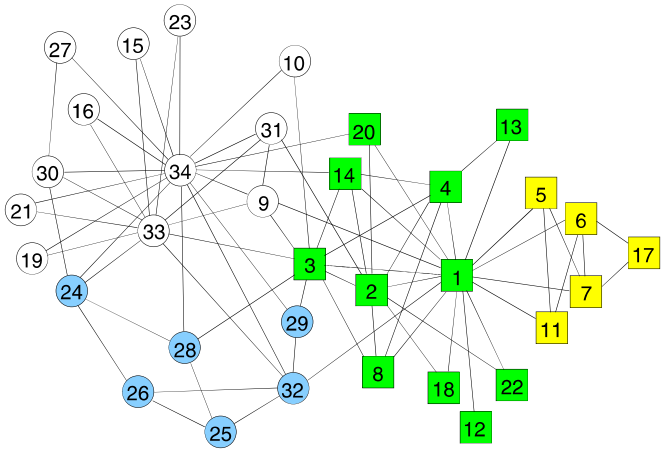




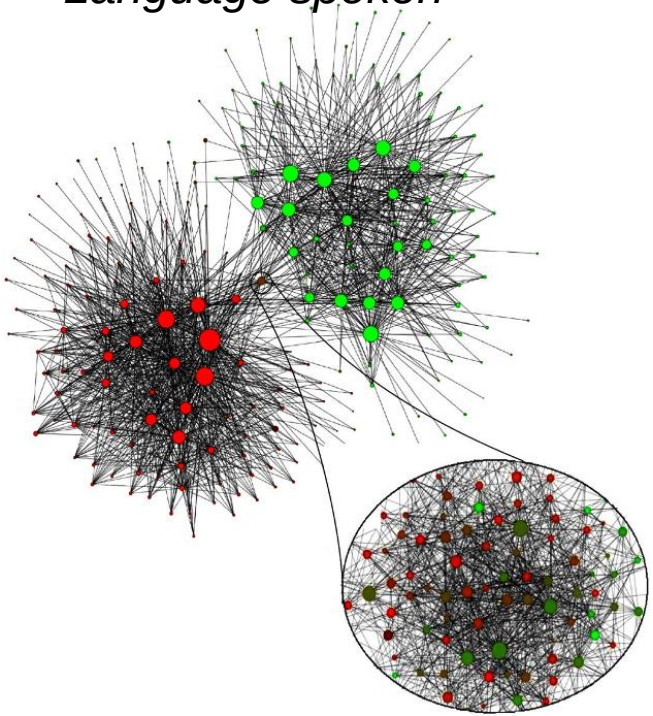
Section 1 Auxiliary information

→ *Karate Club:*
Breakup of the club

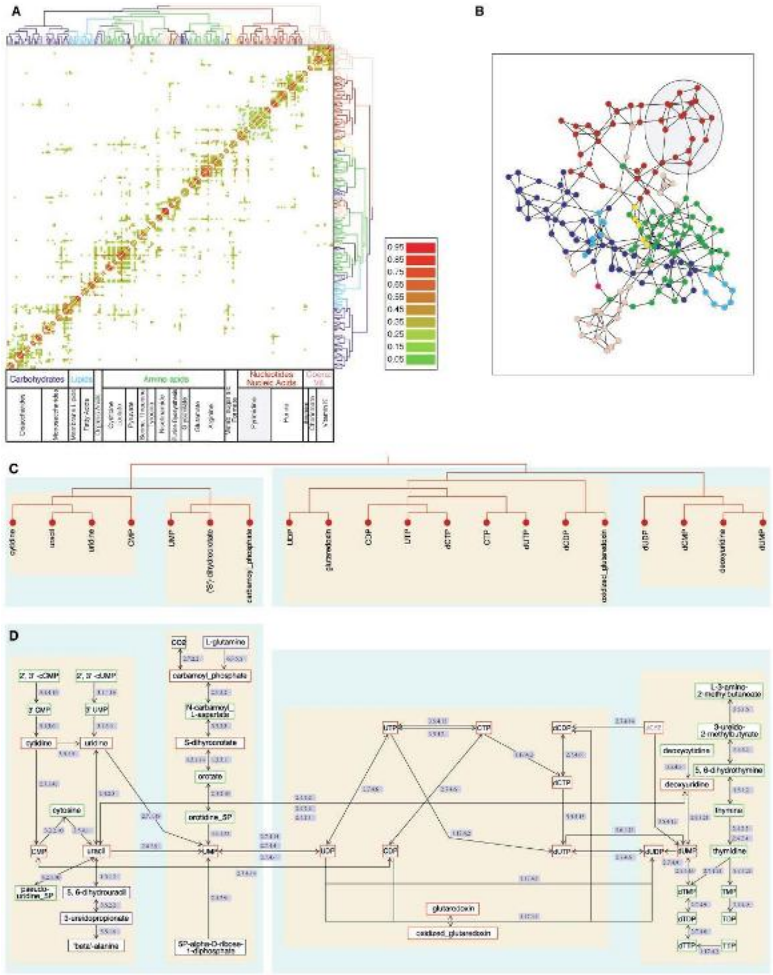
a

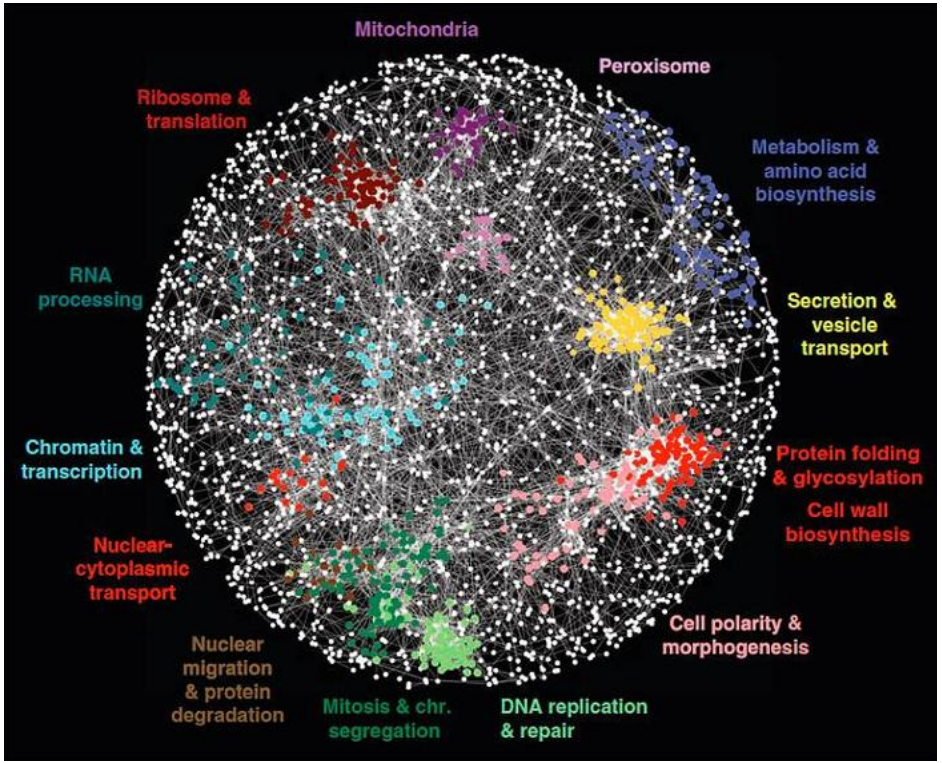


→ *Belgian Phone Data:*
Language spoken



Section 2 Biological Modules





Basics of communities

(i) <u>*Connectedness Hypothesis*:</u>

A *community corresponds to a connected subgraph*.

<u>(ii) *Density Hypothesis*:</u>

Communities correspond to locally dense neighborhoods of a network.

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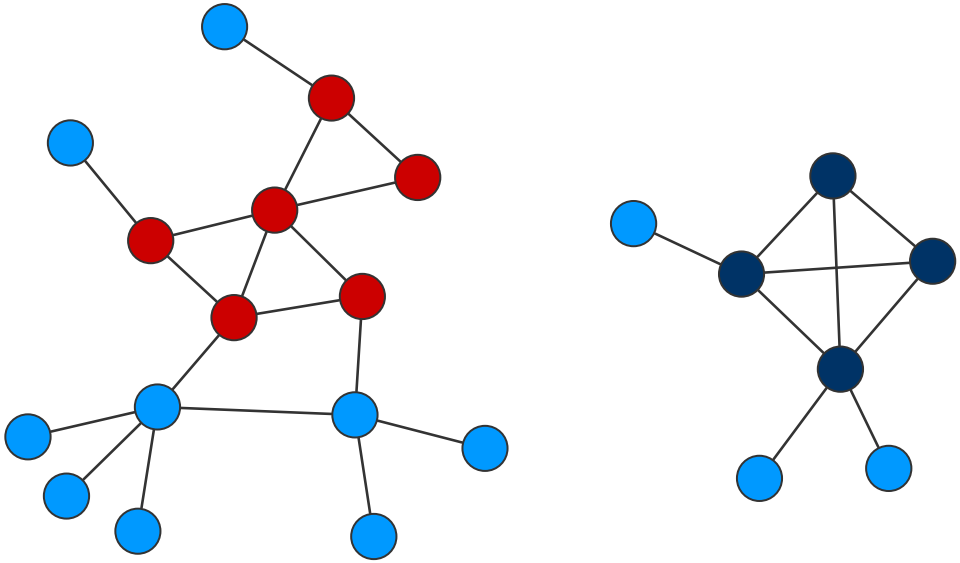


Internal degree, k^{int} : set of links that connect to other nodes of the same community.

External degree k^{ext}: the set of links that connect to the rest of the network.

If $k^{ext}=0$: a community is a good one for node.
If $k^{int}=0$, then the node must be assigned to a different community.

intra-cluster density

$$d^{int}(C) = \frac{\text{Number of internal links of C}}{N_C(N_C - 1)/2}$$

inter-cluster density

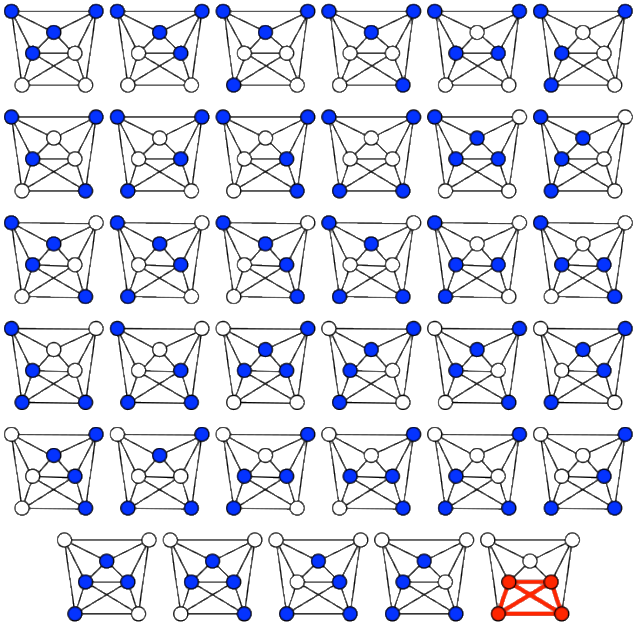
$$d^{ext}(C) = \frac{\text{Number of intercluster links of C}}{N_C(N - N_C)}$$

$$\delta^{\text{int}}(C) \gg \langle k \rangle \gg \delta^{\text{ext}}(C)$$

Community definitions (1)

Maximum Cliques: $\delta^{\text{int}}(C) = 1$

- Triangles are frequent; larger cliques are rare.
- Communities do not necessarily correspond to complete subgraphs, as many of their nodes do not link directly to each other.
- Finding the cliques of a network is computationally rather demanding, being a so-called NP-complete problem.



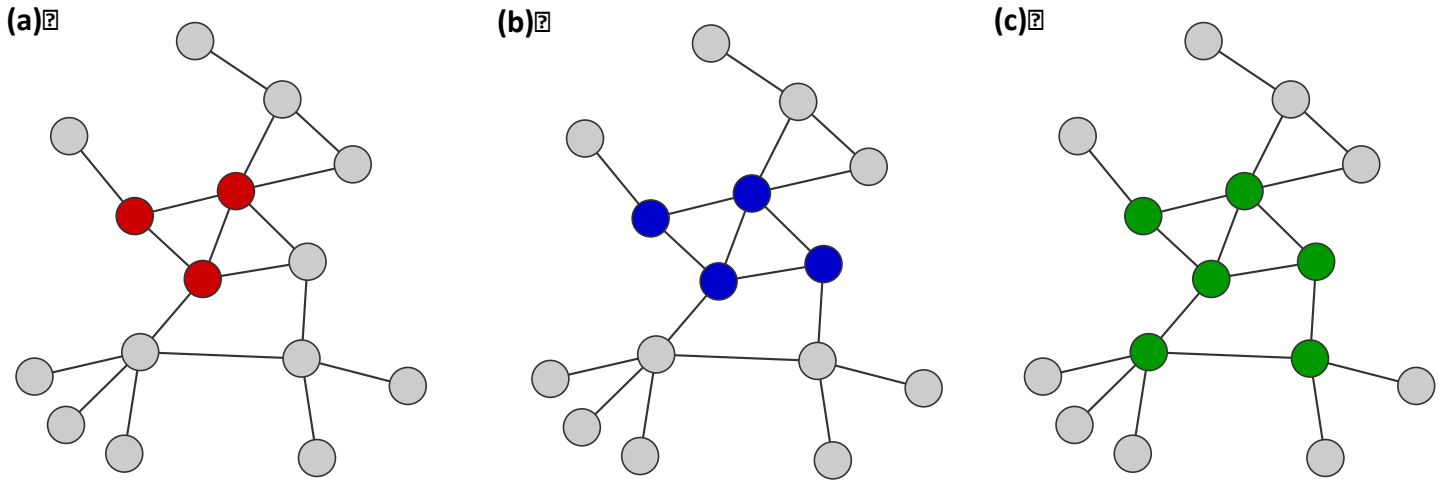
Community definitions (2)

strong community:
each node has more links within the community than with the rest of the graph.

$$k_i^{\mathrm{int}}(C) > k_{\mathrm{i}}^{\mathrm{ext}}(C)$$

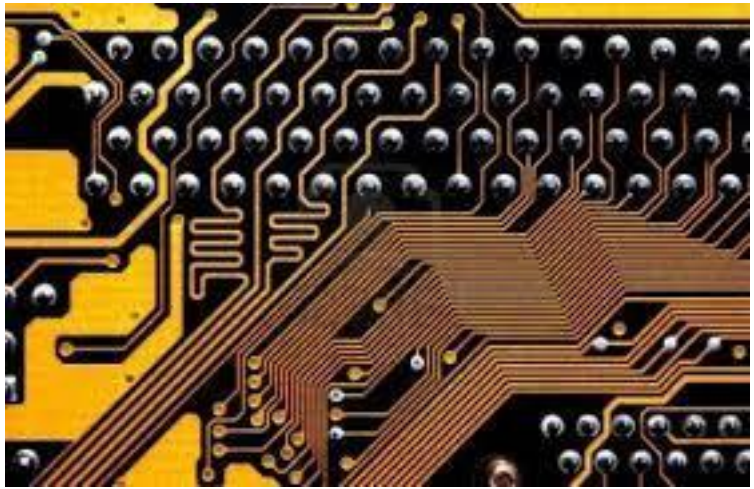
weak community:
the total internal degree of the subgraph exceeds its total external degree,

$$\sum_{\mathrm{i}\in\mathrm{C}} k_i^{\mathrm{in}}(C) > \sum_{\mathrm{i}\in\mathrm{C}} k_{\mathrm{i}}^{\mathrm{out}}(C)$$



Graph Partitions (history)

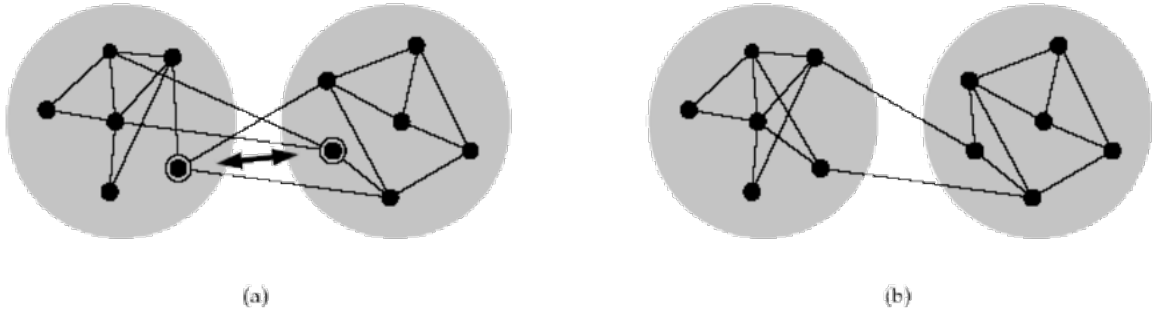
2.5 billion transistors



graph partitioning

partition the full wiring diagram of an IC into smaller subgraphs, so that they minimize the number of connections between them.

Kerninghan-Lin Algorithm



- Partition a network into two groups of predefined size.
- Inspect each a pair of nodes, one from each group. Identify the pair that results in the largest reduction of the cut size if we swap them
- Swap them, like the pair highlighted in (a), arriving to the partition shown in (b). If no pair reduces the cut size, we swap the pair that increases the cut size the least.
- The process is repeated until each node is moved once.

How many ways can we partition a network into communities?

graph bisection

divide a network into two equal non-overlapping subgraphs, such that the number of links between the nodes in the two groups is minimized.

Two subgroups of sized n_1 and n_2: $$\frac{N!}{n_1!n_2!}$$

$$n! \simeq \sqrt{2\pi n}(n/e)^n$$

$$\frac{N!}{n_1!n_2!} \simeq \frac{\sqrt{2\pi N}(N/e)^n}{\sqrt{2\pi n_1}(n_1/e)^{n_1}\sqrt{2\pi n_2}(n_2/e)^{n_2}} \simeq \frac{N^{N+1/2}}{n_1^{n_1+1/2}n_2^{n_2+1/2}}$$

$$n_1 = n_2 = N/2$$

$$\frac{2^{N+1}}{\sqrt{N}} = e^{(N+1)\ln 2/\sqrt{N}}$$

N=10 → 256 partitions (1 ms)

N=100→ 10^{26} partitions (10^{20} years)

Number of communities

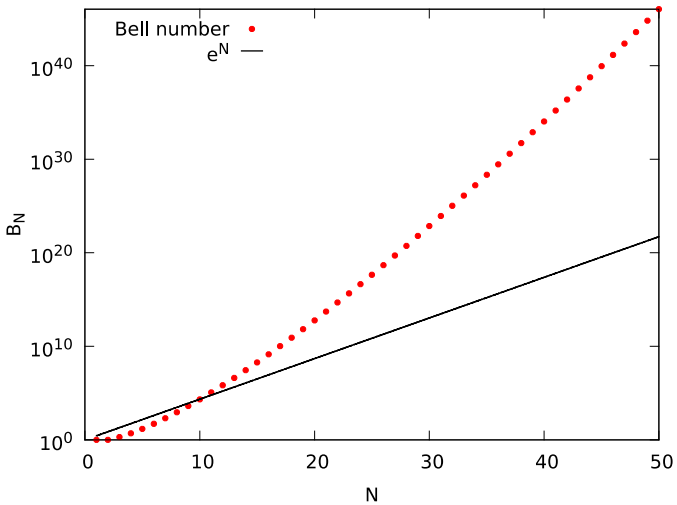
community detection:
the number and size of the communities are unknown at the beginning.

Partition:
division of a network into groups of nodes, so that each node belongs to one group.

$$B_N = \sum_{k=0}^{N} S(N,k)$$

$$B_N \approx \frac{1}{\sqrt{N}} \left[l(N) \right]^{N+\frac{1}{2}} e^{l(N) - N - 1}$$

$$l(N) = e^{W(N)}$$



Hierarchical Clustering

- *similarity matrix*: how similar two nodes are to each other → we need to determine from the adjacency matrix
- hierarchical clustering iteratively identifies groups of nodes with high similarity, following one of two distinct strategies:

Agglomerative algorithms merge nodes and communities with high similarity.

Divisive algorithms split communities by removing links that connect nodes with low similarity.

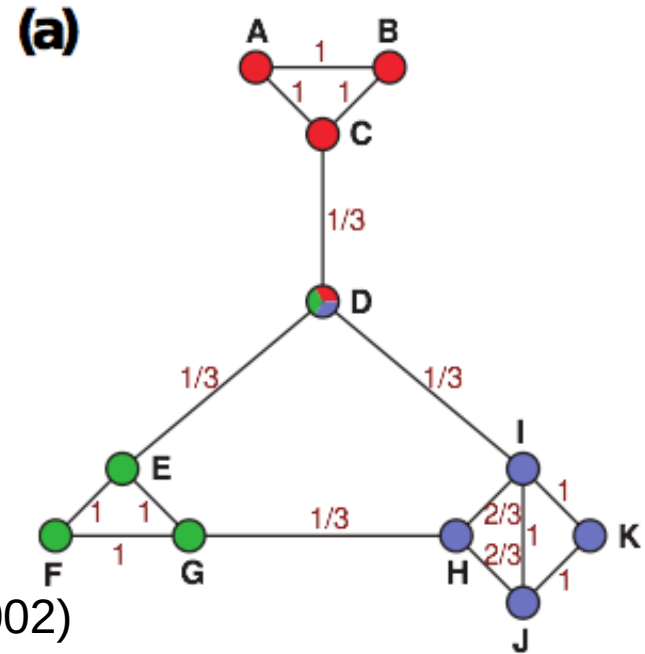
- *hierarchical tree*, or *dendrogram*: visualize the history of the merging or splitting process the algorithm follows. Horizontal cuts of this tree offer various community partitions.

Step 1: Define the Similarity Matrix.

- high for node pairs that likely belong to the same community;
- low for those that likely belong to different communities.
- Nodes that connect directly to each other and/or share multiple neighbors are more likely to belong to the same dense local neighborhood, hence their should be large.

Topological overlap matrix: $$x_{ij}^{o} = \frac{J_n(i,j)}{\min(k_i, k_j)}$$

$J_N(i,j)$: number of common neighbors of node i and j;
(+1) if there is a direct link between i and j;

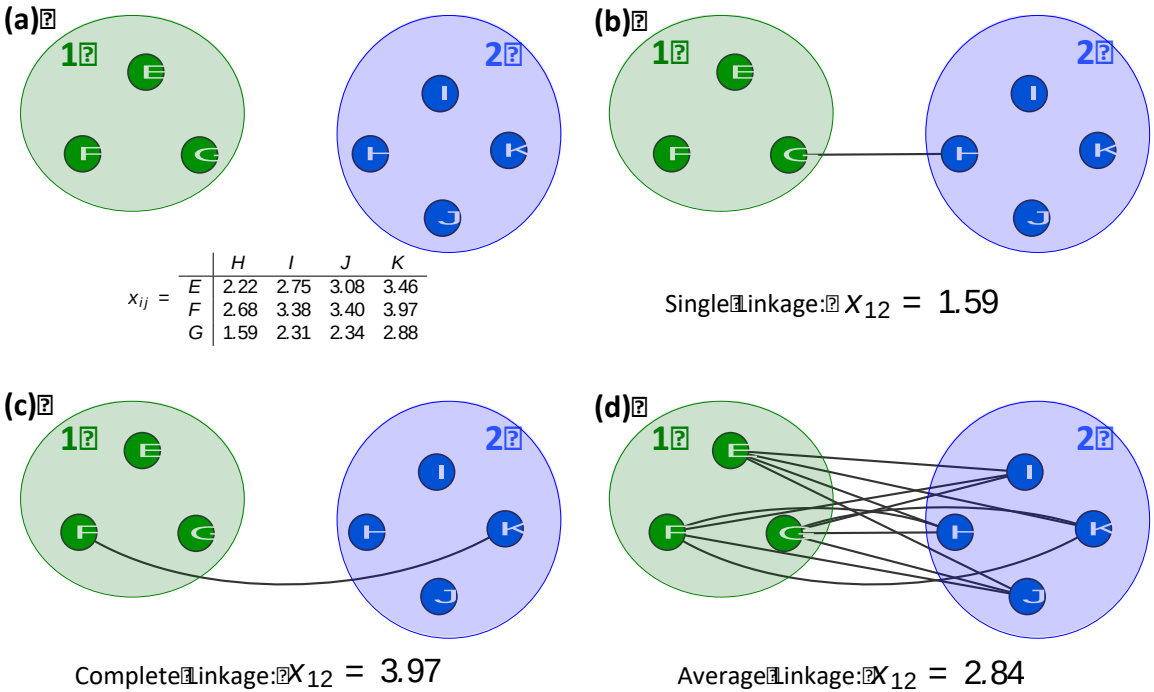


Ravasz, Somera, Mongru, Oltvai Barabási. *Science* (2002)

Step 2: Decide Group Similarity.

- groups are merged based on their mutual similarity:

single, *complete* and *average cluster similarity*



x_{ij} =

	H	I	J	K
E	2.22	2.75	3.08	3.46
F	2.68	3.38	3.40	3.97
G	1.59	2.31	2.34	2.88

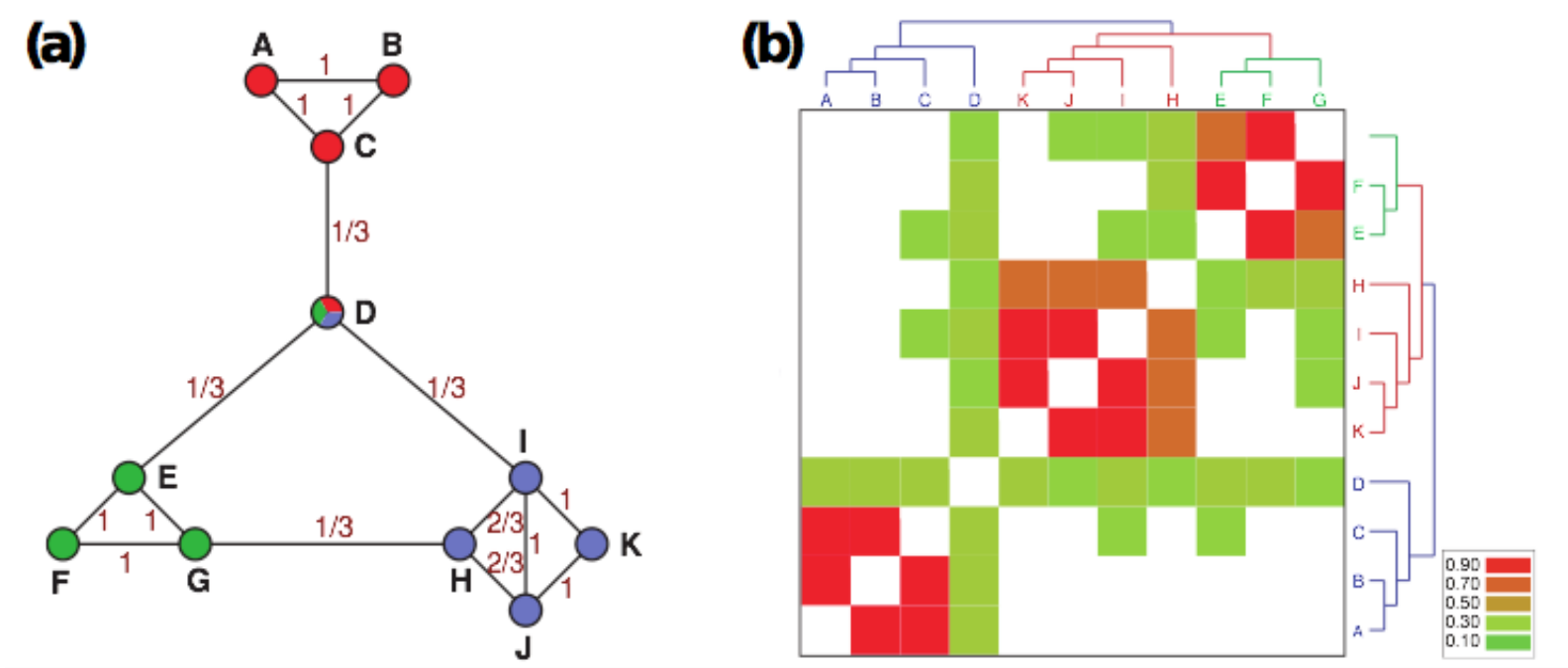
Step 3: Apply Hierarchical Clustering

- Assign each node to a community of its own and evaluate for all node pairs. The initial similarities between these “communities” are simply the node similarities.
- Find the community pair with the highest similarity and merge them to form a single community.
- Calculate the similarity between the new community and all other communities.
- Repeat from Step 2 until all nodes are merged into a single community.

Step 4: Build Dendrogram.

- describes the precise order in which the nodes are assigned to communities.

Agglomerative Algorithms

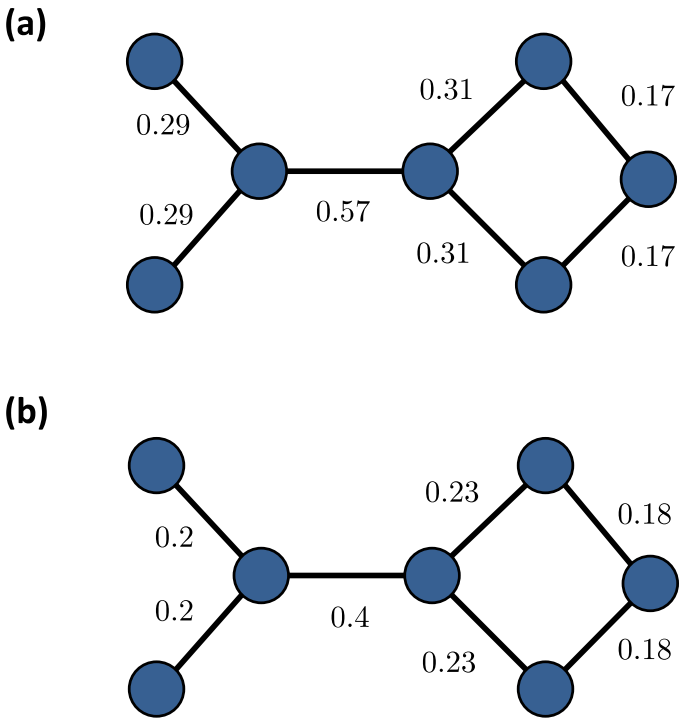


Divisive algorithms

systematically remove the links that connect nodes that belong to different communities, allowing the network fall into isolated clusters.

Step 1: Define a Centrality Measure:

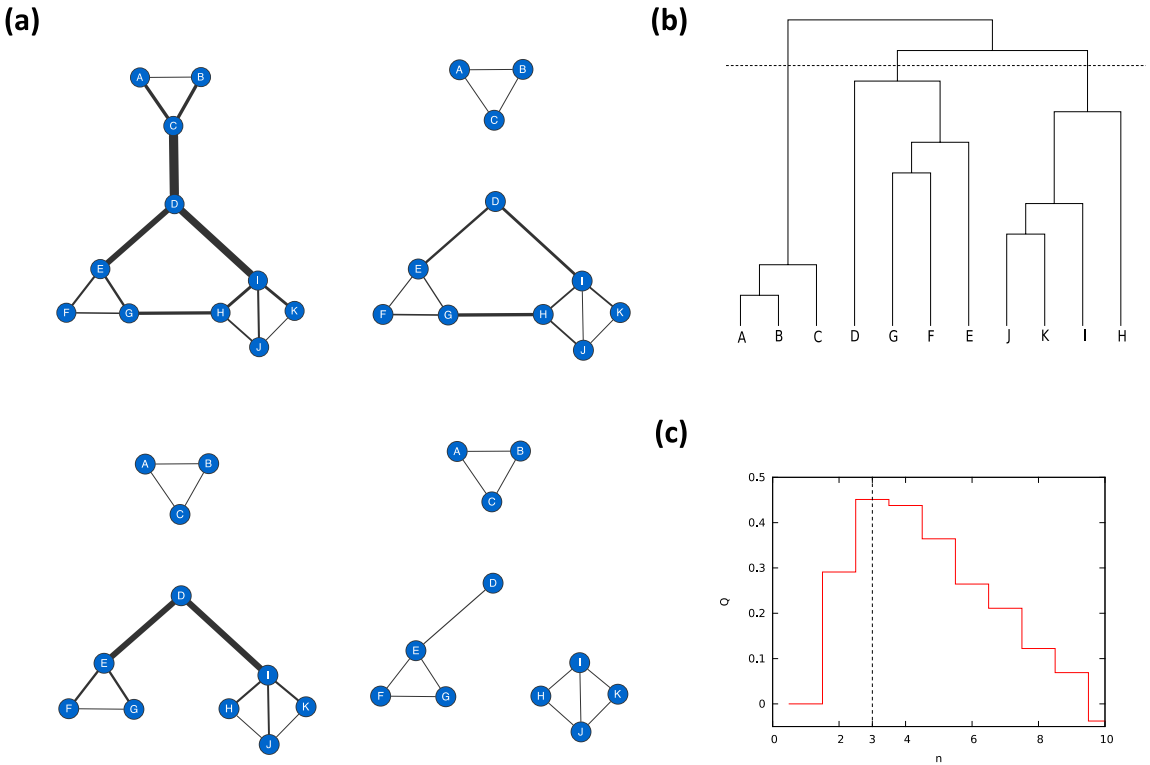
- *Link betweenness* is the number of shortest paths between all node pairs that run along a link.
- *Random-walk link betweenness*. A pair of nodes and is chosen at random, and a walker starts at , following each adjacent link with equal probability until it reaches
- .
- *Current-flow betweenness* assumes that each link is a resistor with unit resistance.



M. Girvan and M.E.J. Newman. PNAS 2002.

Step 2: Hierarchical Clustering:

- Compute of the centrality of each link.
- Remove the link with the largest centrality; in case of a tie, choose one randomly.
- Recalculate the centrality of each link.
- Repeat from step 2 until all links are removed.



- *Hierarchy Assumption*:

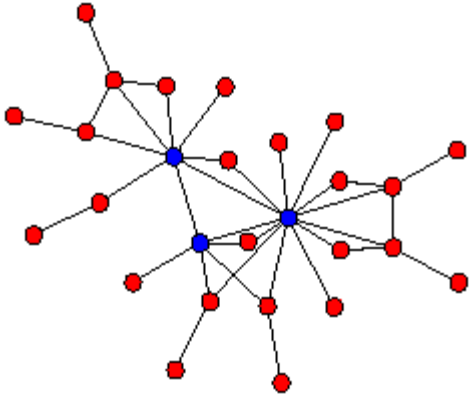
communities are organized in a hierarchical fashion. This hierarchy is captured by the dendrogram, telling how the small modules are nested into larger ones.

- *Relation to scale-free networks*:

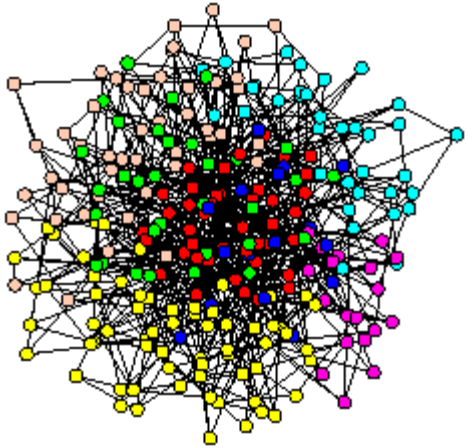
Community identification assumes that a network can be partitioned into a collection of subgraphs that are locally dense and are only sparsely connected to other communities. Yet, the scale-free property implies the existence of hubs, that given the many links they have, must connect to nodes that belong to different communities

Hierarchy in networks

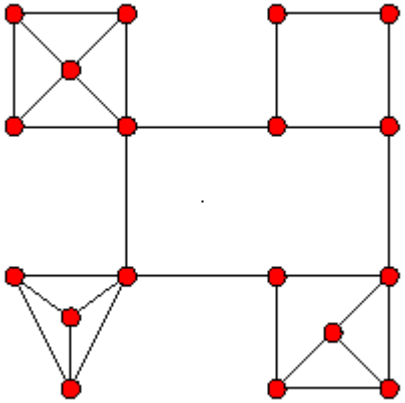
(a)



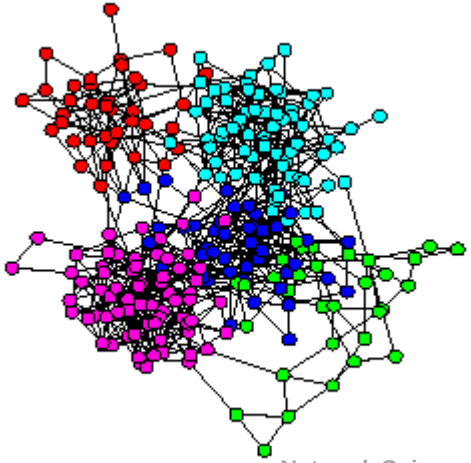
Scale-free



(b)

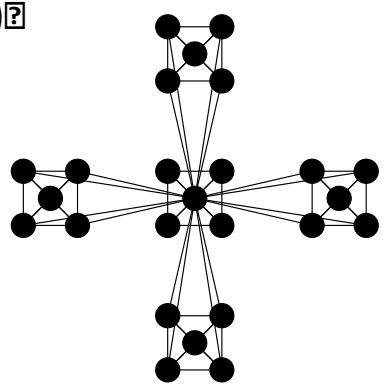


Modular

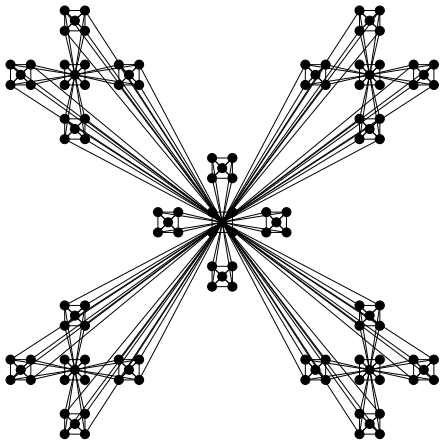


(a)

(b)



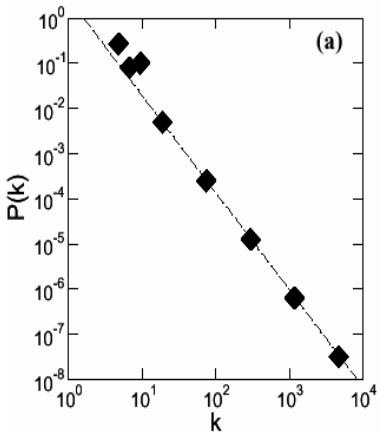
(c)



(1) Scale-free property

The obtained network is scale-free, its degree distribution following a power-law with

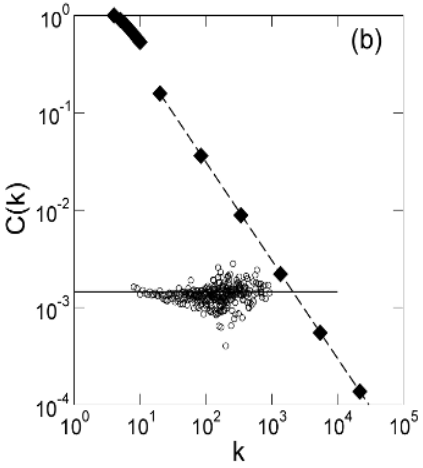
$$\gamma = 1 + \frac{\ln 5}{\ln 4} \simeq 2.16$$



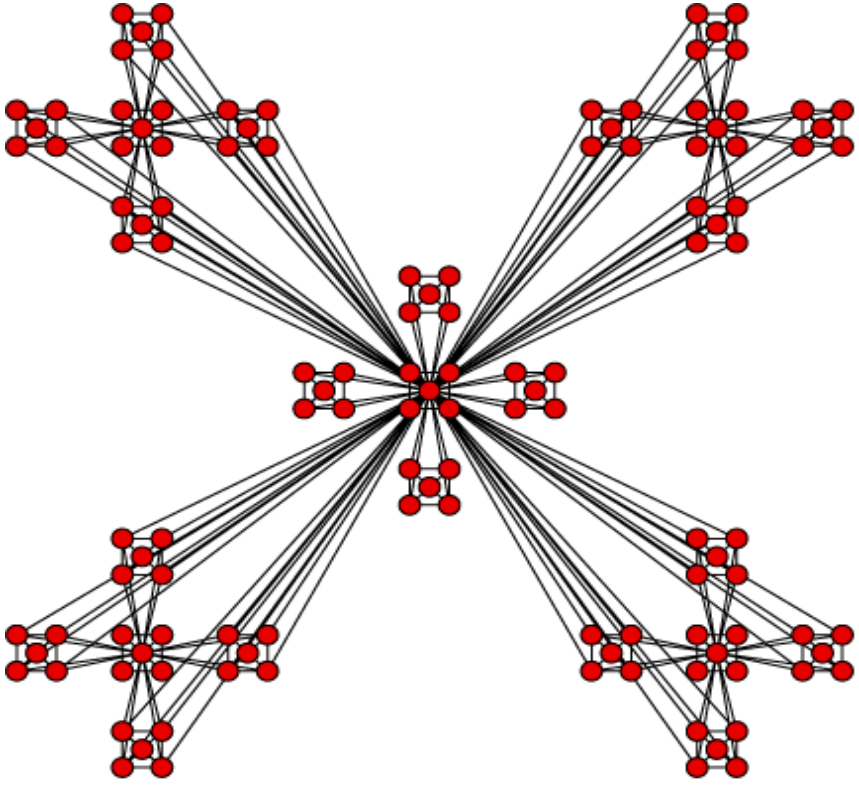
HIERARCHICAL NETWORKS

(2) Clustering coefficient

$$C(k) \sim k^{-1}$$



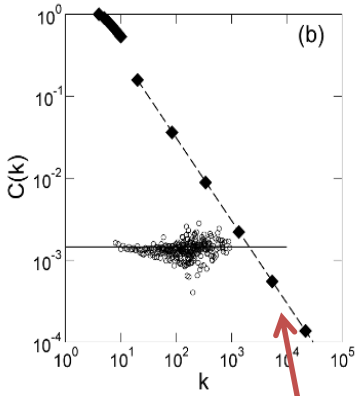
$$C(k) = \frac{\text{\# links between k neighbors}}{k(k-1)/2}$$

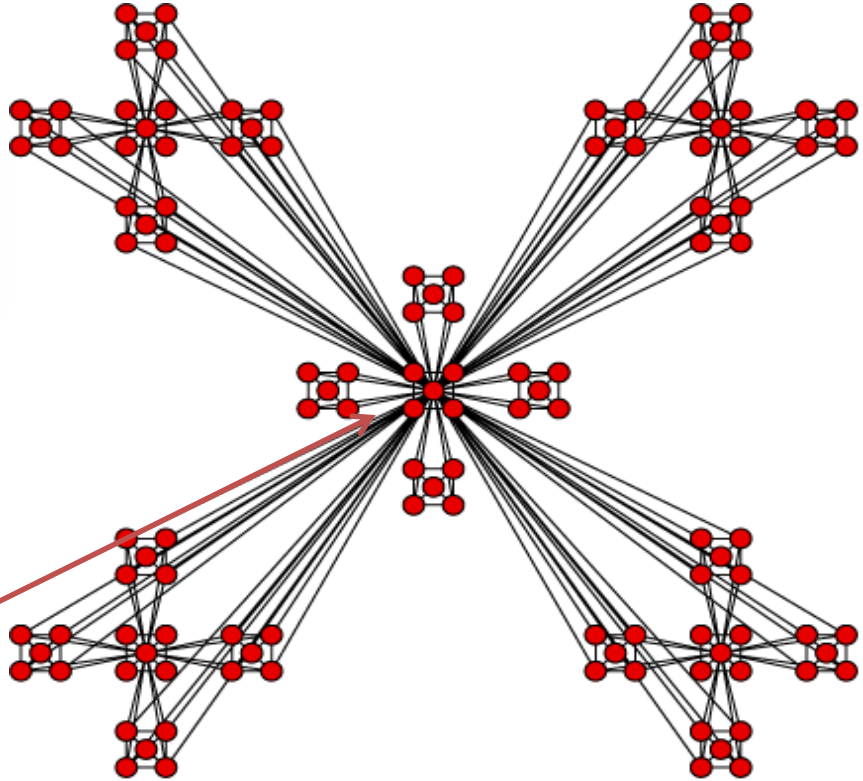


WHAT DOES THE SCALING MEAN?

$$C(k) \sim k^{-1}$$

Small *k* nodes:
high clustering coefficient;
their neighbors tend to link to each other;
in highly interlinked, compact communities.



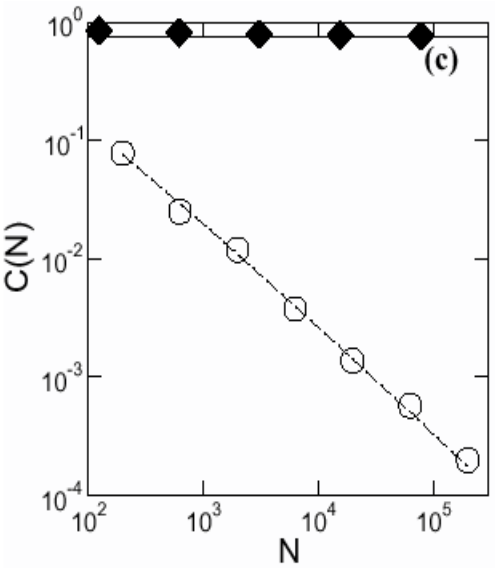


High *k* nodes (hubs):
*small clustering coefficient;
*connect independent communities.

PROPERTIES OF THE MODEL

Large average clustering coefficient

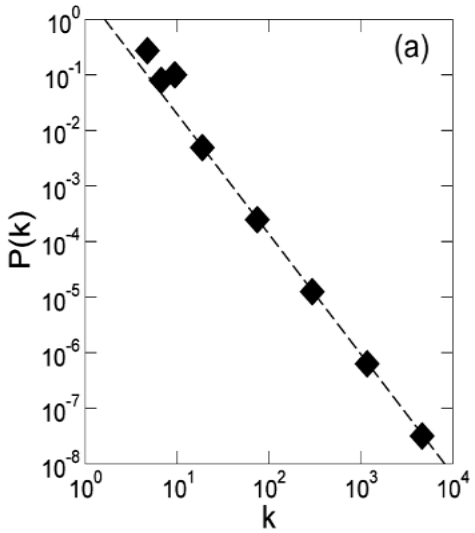
$$C \simeq 0.74$$



PROPERTIES OF HIERARCHICAL NETWORKS

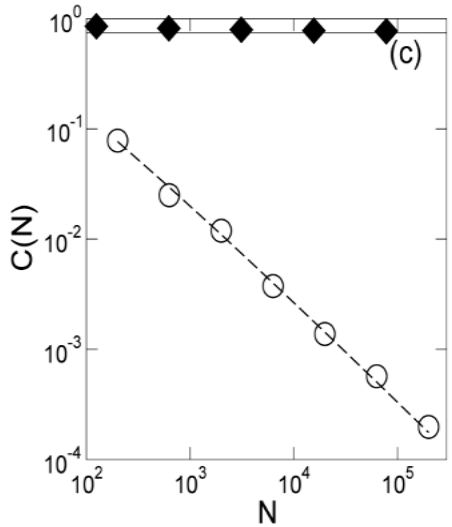
1. Scale-free

$$\gamma = 1 + \frac{\ln 5}{\ln 4} = 2.161$$



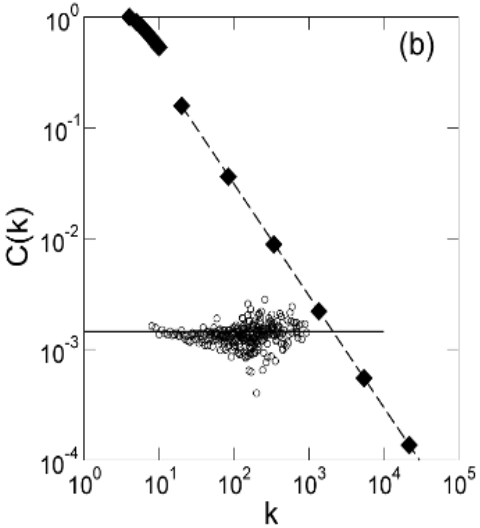
2. Clustering coefficient independent of N

$$C(N) = \text{const.}$$



3. Scaling clustering coefficient (DGM)

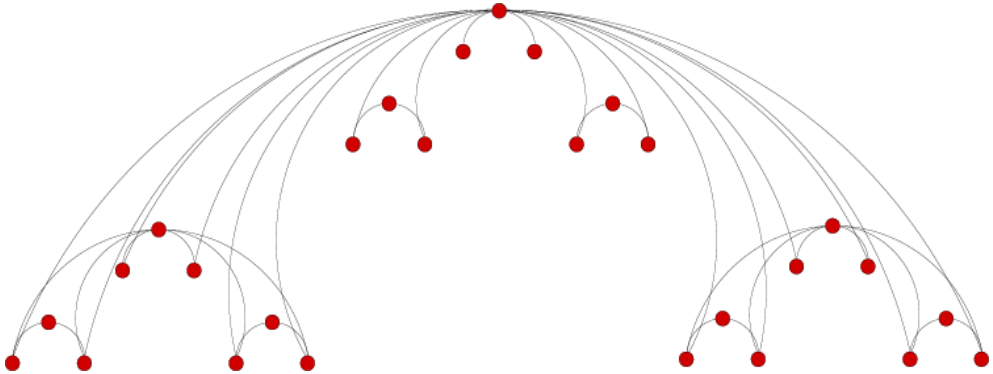
$$C(k) \sim k^{-1}$$



HIERARCHICAL MODELS

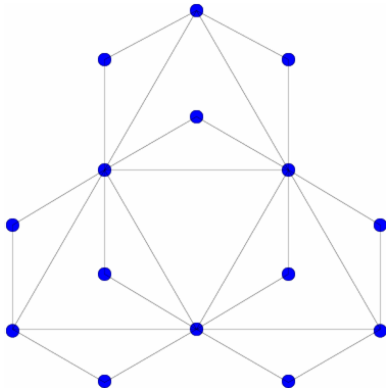
$$g = \frac{\ln 3}{\ln 2}$$

- Barabási, Ravasz, Vicsek, Physica A 2003



- Dorogovtsev, Goltsev, Mendes, 2001

$$g = 1 + \frac{\ln 3}{\ln 2}$$



Modularity

***Random Hypothesis*:**
Randomly wired networks are not expected to have a community structure.

$$\{C_c,\ c = 1,\ n_c\}$$

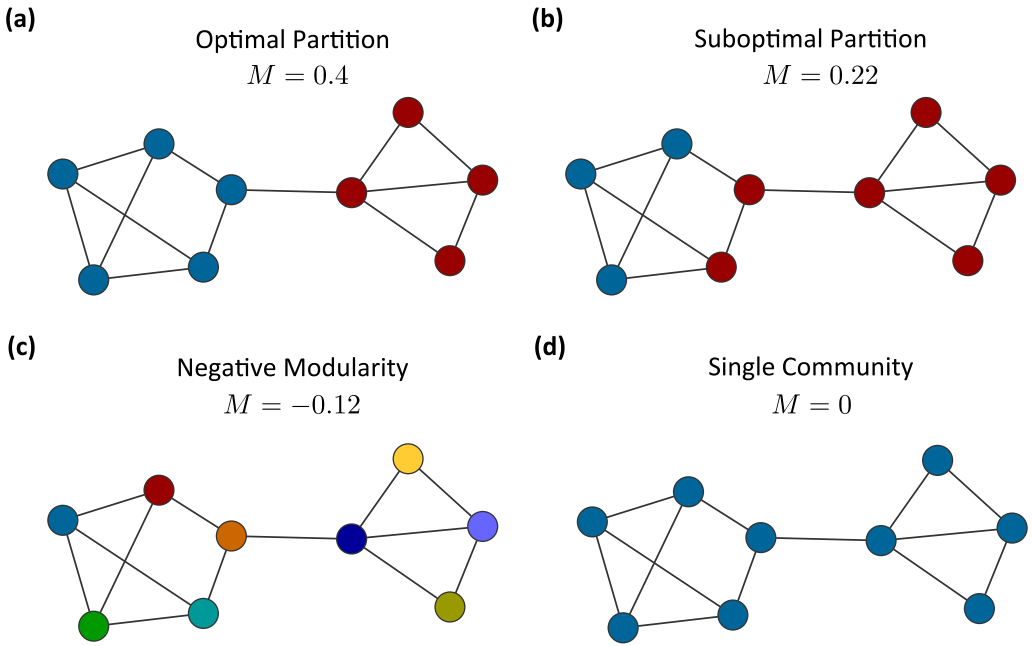
$$M = \frac{1}{2L}\sum_{i,j=1}^{N}(A_{ij} - P_{ij})\delta(C_i - C_j)$$

$$P_{ij} = 2Lp_ip_j = \frac{k_ik_j}{2L}$$

$$M = \sum_{c=1}^{n_c}\left[\frac{l_c}{L} - \left(\frac{k_c}{2L}\right)^2\right]$$

Section 5 Modularity

$$M = \sum_{c=1}^{n_c} \left[\frac{l_c}{L} - \left(\frac{k_c}{2L} \right)^2 \right]$$



- *Optimal partition*, that maximizes the modularity.
- *Sub-optimal* but positive modularity.
- *Negative Modularity*: If we assign each node to a different community.
- *Zero modularity:* Assigning all nodes to the same community, we obtain , independent of the network structure.
- *Modularity is size dependent.*

Modularity based community identification

*Greedy techniques [*Newman], iteratively joins nodes if the move increases the partition's modularity.

1. Assign each node to a community of its own, e.g. start with "communities".

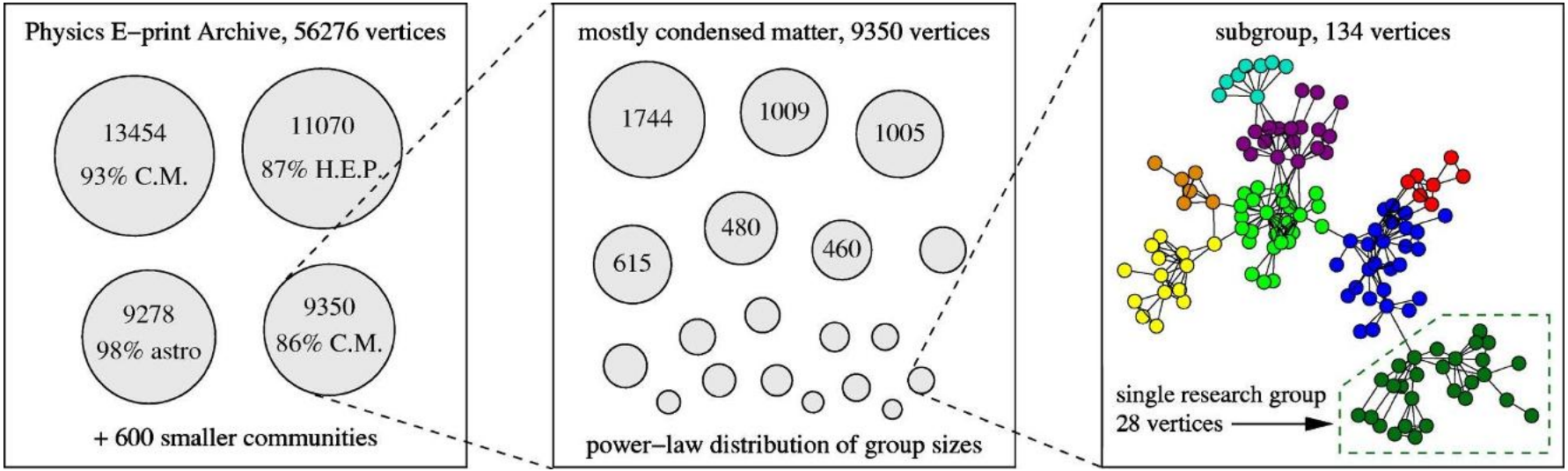
2. Inspect each pair of communities connected by at least one link and compute the modularity variation obtained if we merge these two communities.

3. Identify the community pairs for which ΔM is the largest and merge them. Note that modularity of a particular partition is always calculated from the full topology of the network.

4. Repeat Step 2 until all nodes are merged into a single community.

5. Record for each step and select the partition for which is maximal.

Modularity based community identification

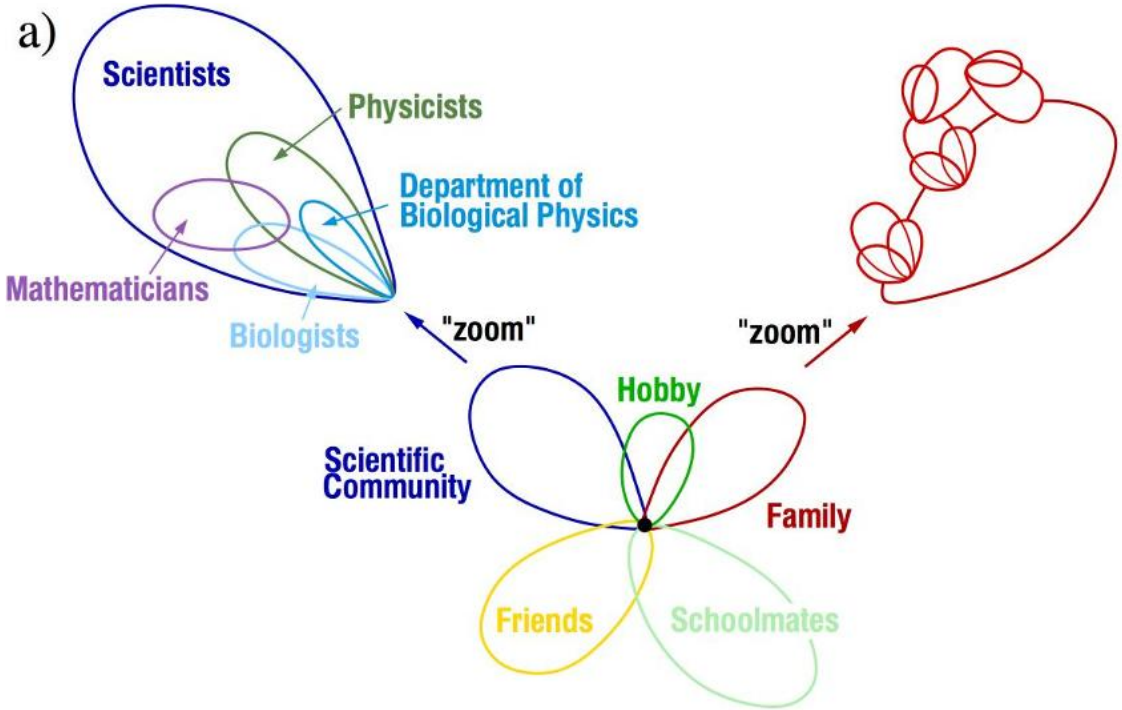


Online Resources (1)

The greedy algorithm is neither particularly fast nor particularly successful at maximizing M.
Scalability: Due to the sparsity of the adjacency matrix, the update of the matrix involves a large number of useless operations. The use of data structures for sparse matrices can decrease the complexity of the computational algorithm to , which allows us to analyze is of networks up to nodes. See
"Fast Modularity" Community Structure Inference Algorithm
http://cs.unm.edu/~aaron/research/fastmodularity.htm for the code.

A fast greedy algorithm was proposed by Blondel and collaborators, that can process networks with millions of nodes. For the description of the algorithm see
Louvain method: Finding communities in large networks
https://sites.google.com/site/findcommunities/ for the code.

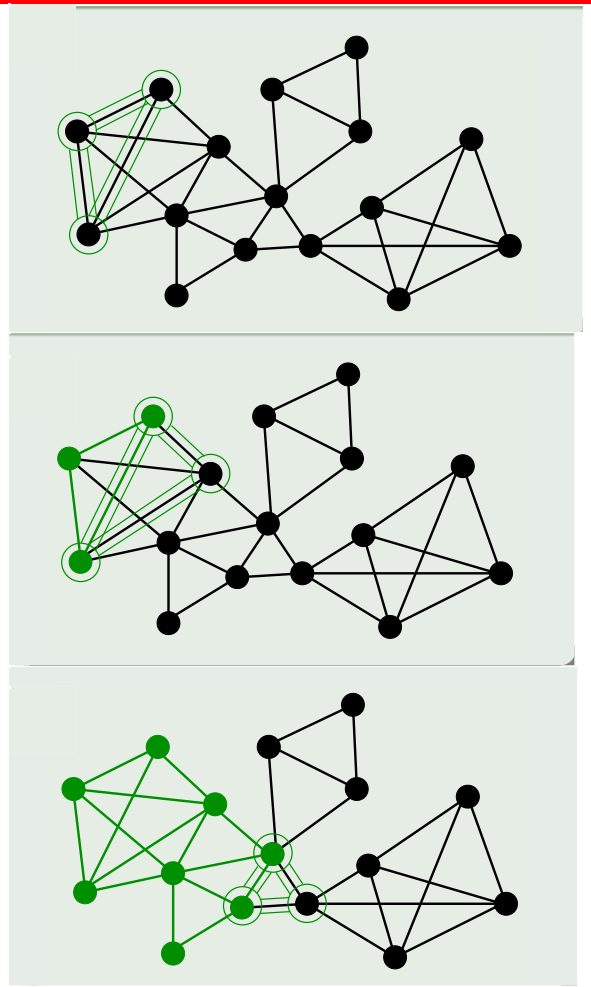
Overlapping Communities



Palla, Derenyi, Farkas, Vicsek. Nature (2005).

Clique Percolation

- Two *k*-cliques (complete subgraphs of *k* nodes) are considered adjacent if they share *k-1* nodes
- A *k*-clique community is the largest connected subgraph obtained by the union of all adjacent *k*–cliques
- Other *k*-cliques that can not be reached from a particular -clique correspond to other -clique-communities



Palla, Derenyi, Farkas, Vicsek. Nature (2005).

Clique Percolation

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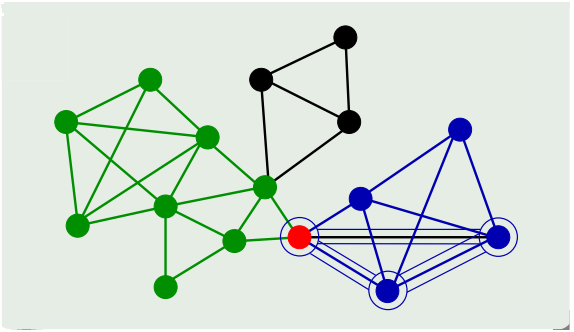
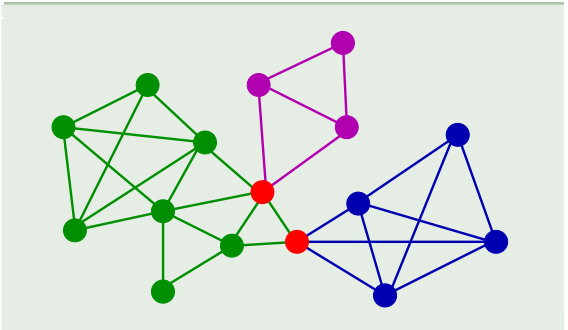
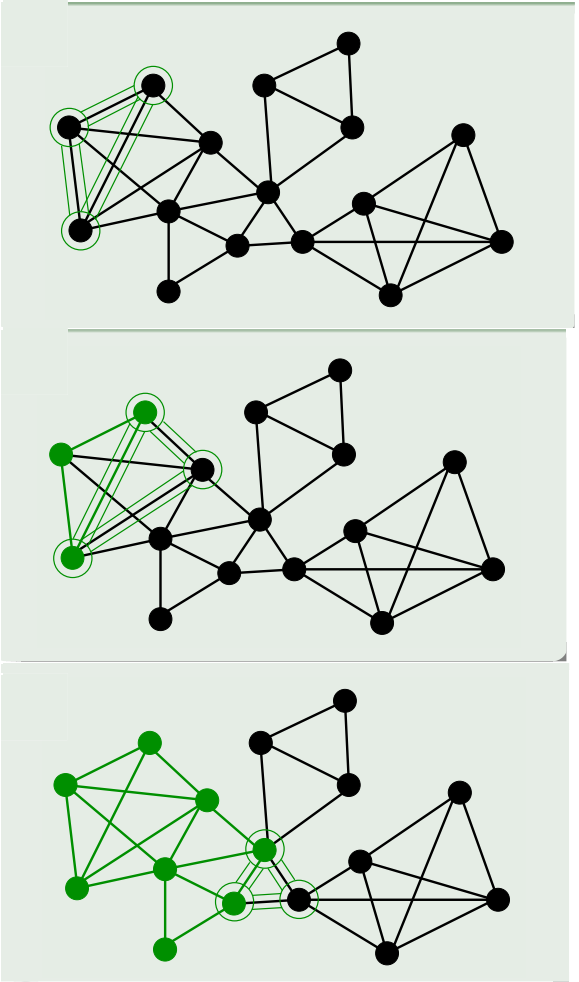
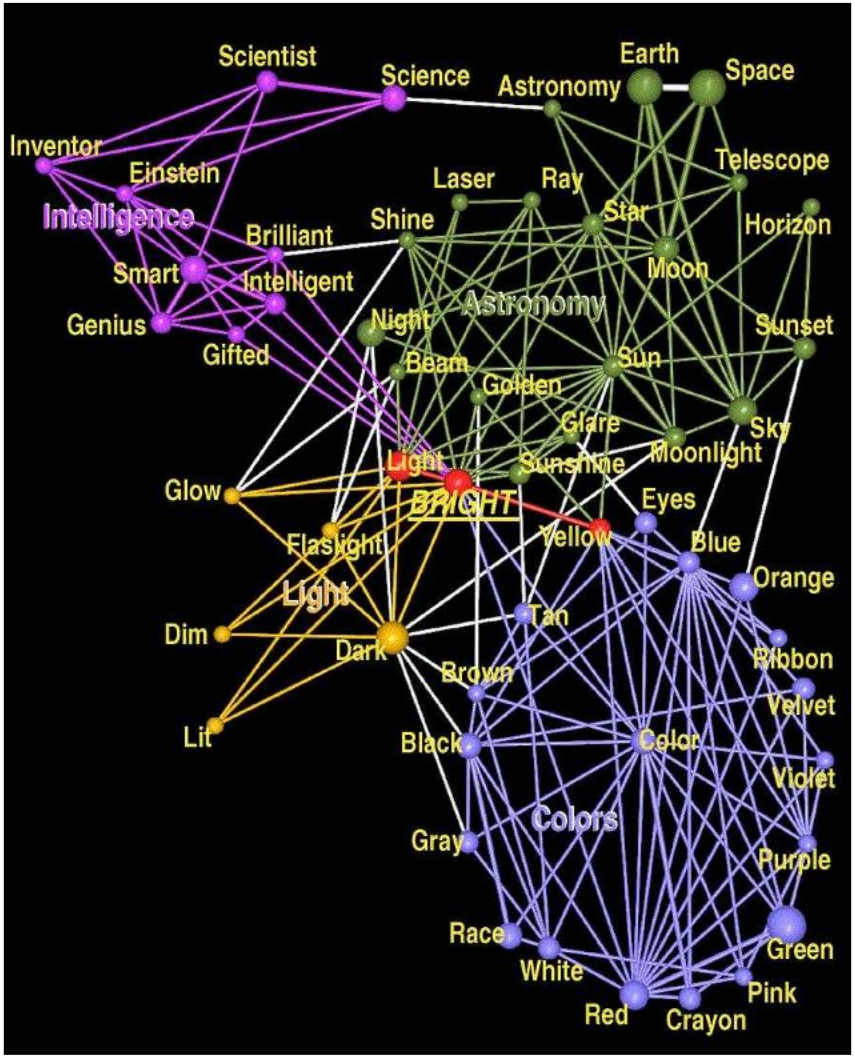






Palla, Derenyi, Farkas, Vicsek. Nature (2005).

Overlapping Communities



bright:

- community containing light-related, *glow* or *dark*;
- community capturing different colors (*yellow*, *brown*)
- community consisting of astronomical terms (*sun*, *ray*).
- community linked to intelligence (*gifted*, *brilliant*).

Could CP communities emerge by chance?

if a random network is sufficiently dense, there are cliques of varying order.

A k-clique community emerges in a random graph only if the connection probability exceeds the threshold:

$$p_c(k) = \frac{1}{[(k-1)N]^{1/(k-1)}}$$

$k = 2$ $\quad p_c(k) \sim 1/N$

$k = 3$ $\quad p_c(k) = 1/\sqrt{2N}$

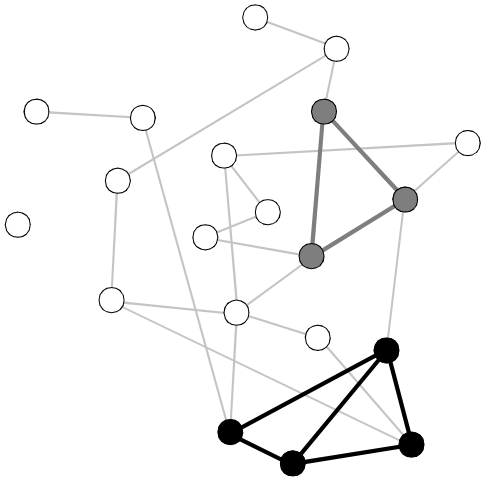
Derényi, Palla, Vicsek. Phys. Rev. Lett. (2005)

Could CP communities emerge by chance?

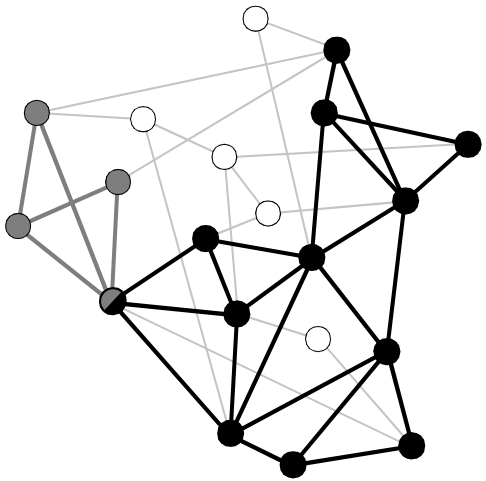
$$p_c(k) = \frac{1}{[(k-1)N]^{1/(k-1)}}$$

$$k = 3 \quad p_c(k) = 1/\sqrt{2N}$$

N=20 p_c=0.16



p=0.13 (<p_c)



p=0.22 (>p_c)

Derényi, Palla, Vicsek. Phys. Rev. Lett. (2005)

Online Resources (2)

The CFinder software package that implements the Clique Percolation Method can be downloaded at

www.cfinder.org.

Link Clustering

While nodes tend to belong to multiple communities, links are more specific, capturing the nature of the relationship between two nodes.

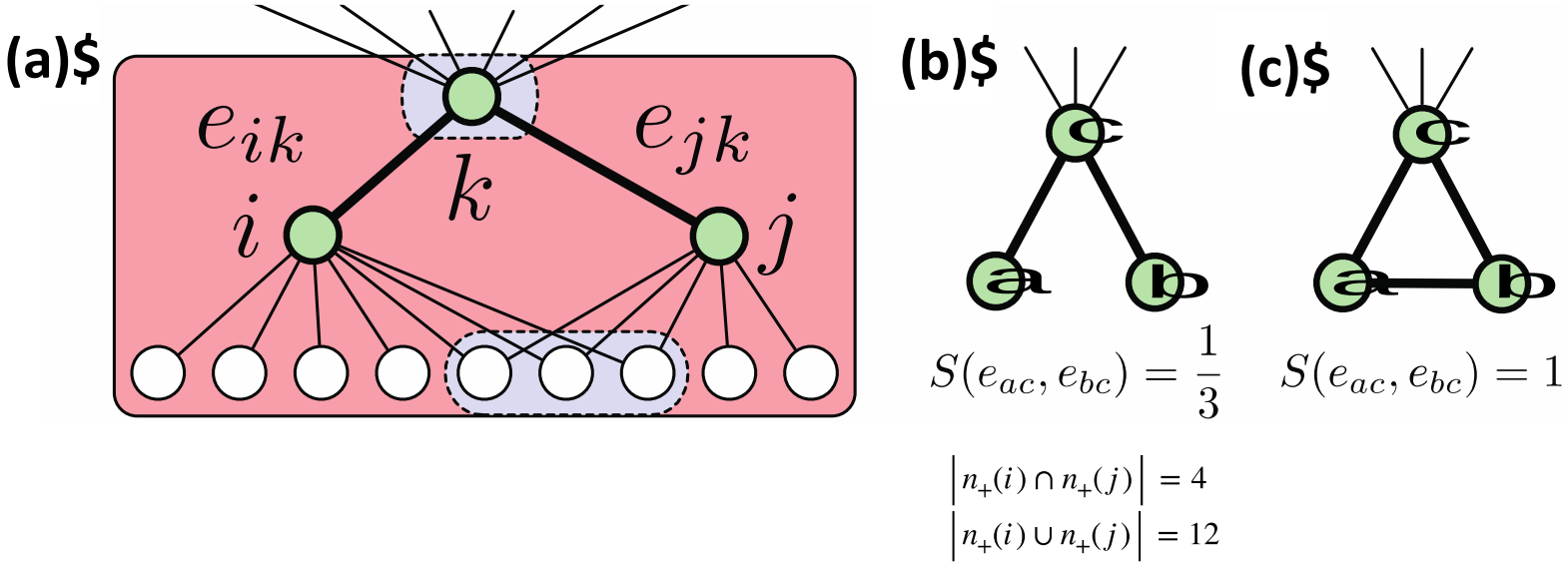
Social networks, a link may indicate:

- they are in the same family
- they work together
- they share a hobby.

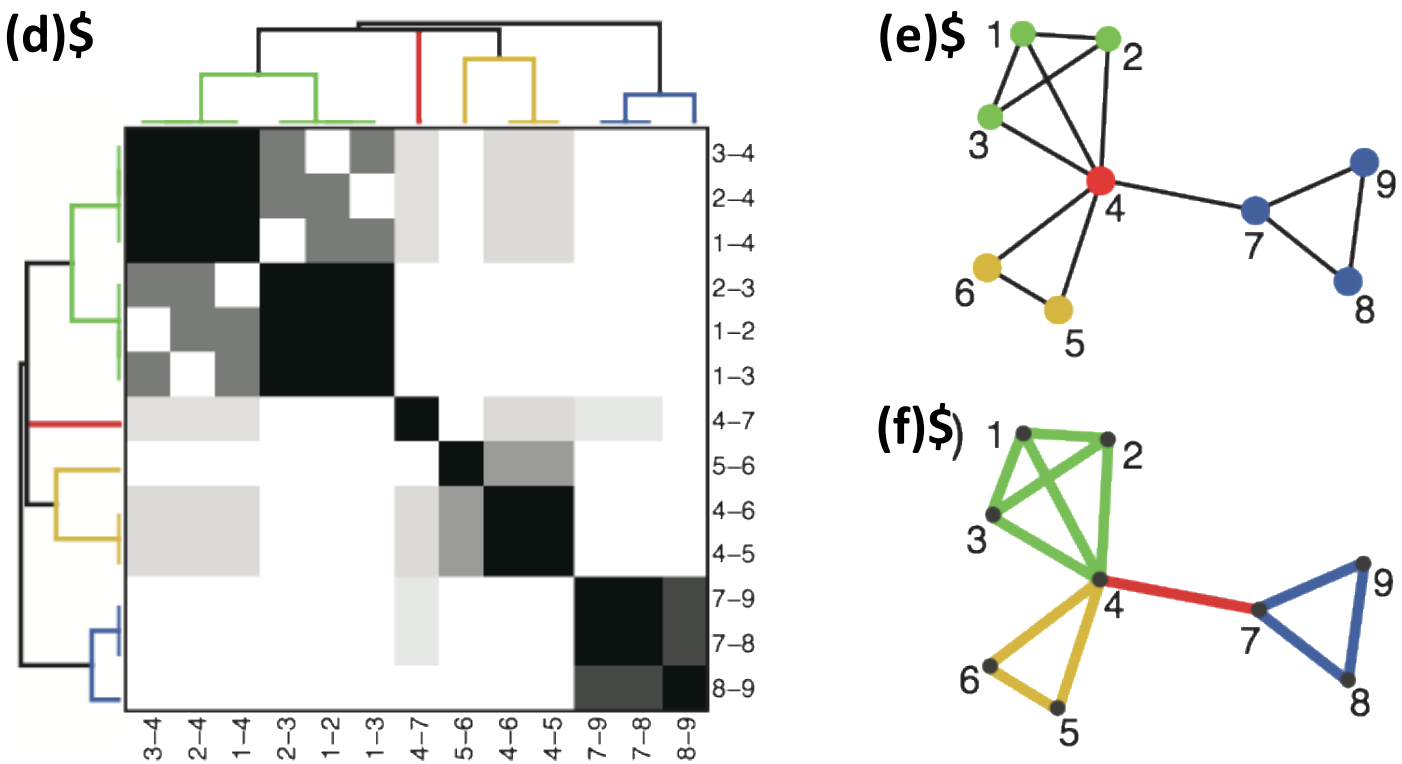
Biological networks:
each interaction of a protein is responsible for a different function, uniquely defining the protein's role in the cell

$$S(e_{ik}, e_{jk}) = \frac{|n_+(i) \cap n_+(j)|}{|n_+(i) \cup n_+(j)|}$$

$n_+(i)$: the list of the neighbors of node i, including itself.
S measures the relative number of common neighbors i and j have.

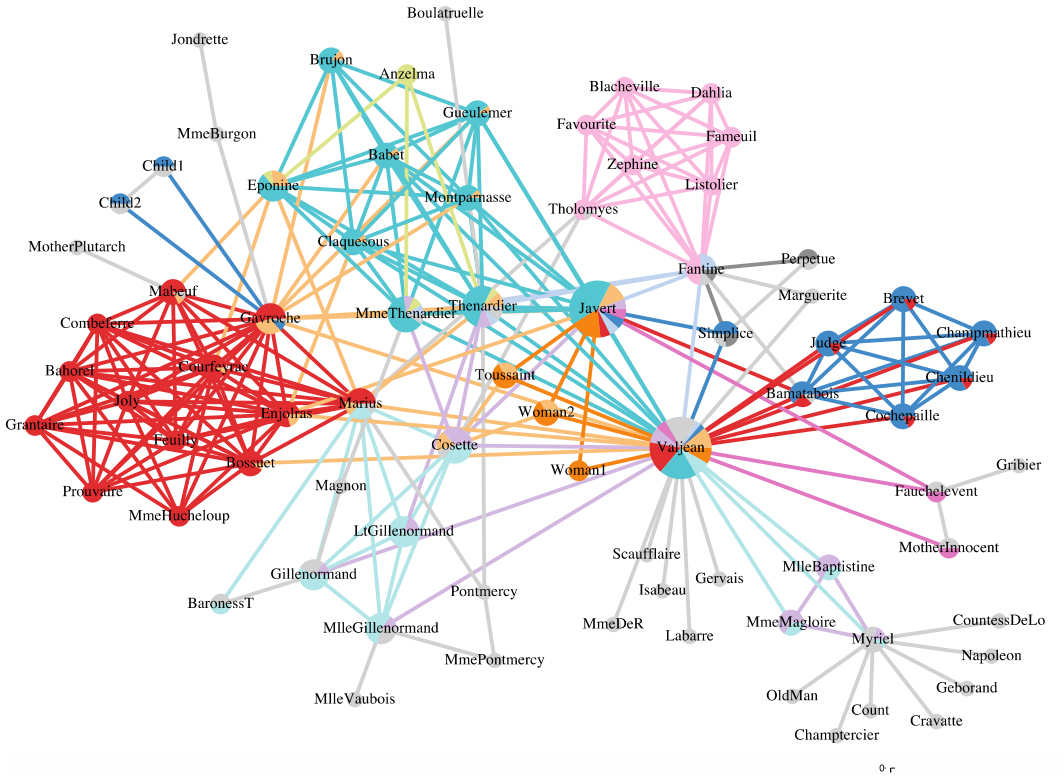


Ahn, Bargow, Lehmann, Nature 2010



Ahn, Bagrow, Lehmann. Nature (2010).

Section 6 Link clustering



Ahn, Bagrow, Lehmann. Nature (2010).

Summary

Summary

Community identification relies on several fundamental hypotheses, pertaining to the nature of communities:

(i) *Connectedness Hypothesis*: A community corresponds to a connected subgraph.

(ii) *Density Hypothesis*: A community corresponds to locally dense neighborhood of a network.

(iii) *Random Hypothesis*: Randomly wired networks are not expected to have a community structure.

(iv) Maximal Modularity Hypothesis: The partition with the maximum modularity offers the best community structure, where modularity is given by

$$M = \sum_{c=1}^{n} \left[\frac{l_c}{L} - \left(\frac{d_c}{2L} \right)^2 \right].$$